Training by the Research Computing Group

SFU SIMON FRASER UNIVERSITY

Digital Research Alliance of Canada

Join our weekly mailing list

https://training.westdri.ca/contact

Training team

MARIE-HÉLÈNE BURLE
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- Background in evolutionary & behavioural ecology (SFU)
- Passionate open-source advocate, certified Software Carpentry instructor
- 18 years training and support experience (university courses, Research / Student Learning Commons, WestGrid, RCG)

ALEX RAZOUMOV
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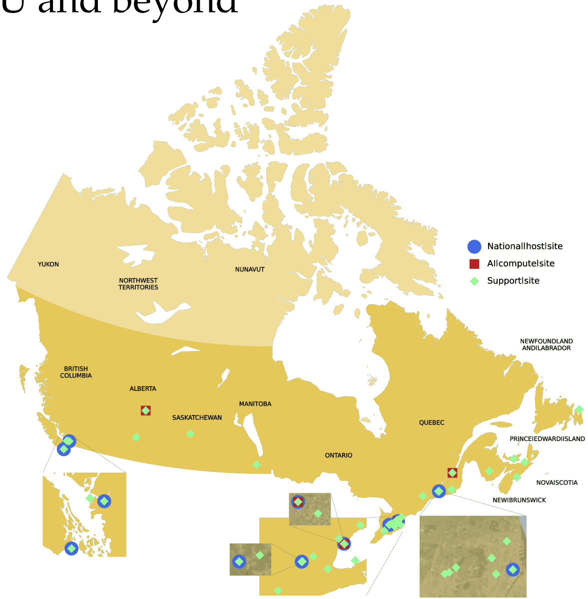
- Background in computational astrophysics (UBC)
- Postdocs in Urbana-Champaign, San Diego, Oak Ridge, & Halifax
- 30+ years training and support experience (university courses, SHARCNET, WestGrid, RCG)
- National Visualization Team lead since 2014, certified Software Carpentry instructor

Both can be reached at training@westdri.ca

Research computing training at SFU and beyond

- Our funding:
  - ✔ supports training researchers in their actual computational workflows, to be run on our systems (max science, max hardware utilization)
  - ✘ does not support undergraduate academic programs (teaching university courses)
- In practice, we often see graduate students lacking basic computational skills, highlighting the need for earlier training and closer collaboration with departments
- SFU is one of the Alliance's national host sites $\Rightarrow$ our training and support are available to all academic researchers across Canada
  - anyone with a valid Canadian university email address can register for our courses
  - webinars and online materials open to anyone
  - we answer national support tickets and contribute to the national docs https://docs.alliancecan.ca



Topics: full-day courses

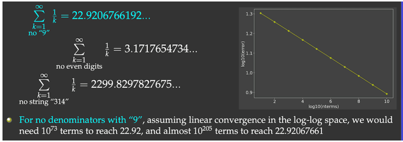
Details/abstracts at https://training.westdri.ca/courses

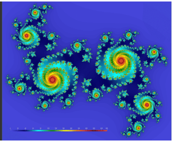
- Remote computing basics
  - Bash command line
  - Introduction to HPC
  - \+ Specific remote computing tools
- Programming basics
  - Python • R • Julia
  - Scientific Python
  - Webscraping or GIS in Python or R
  - Selected Python libraries: Polars, PyTables, ...
- Parallel coding (next slide)
  - Julia • Chapel • Python • R
  - GPU computing with Chapel, Python, Julia
- Deep learning
  - PyTorch • JAX+Flax
- Virtualization
  - Intro to Apptainer containers
  - Intro to cloud VMs
- Scientific visualization (15+ workshops)
  - ParaView • VisIt • 3D vis for the humanities
  - Python/R plotting libraries
  - In-situ vis, remote and large-scale parallel rendering, scripting, animation, interactive web visualization, other advanced sci-vis topics
- RDM topics
  - Version control with Git
  - Distributed datasets: git annex, or DataLad, or DVC
  - Managing files and formats: scientific file formats, parallel I/O, container overlays, etc.
- Publishing
  - R Markdown and Quarto • Interactive dashboards

Parallel coding: our bread and butter

- 10-15 years ago we used to teach MPI (message-passing interface) and OpenMP in C/C++/Fortran (most common HPC languages)
- Today's "Intro to HPC" covers MPI and OpenMP when discussing parallel frameworks
  - but we don't teach actual coding in this course
  - focus on basic cluster tools, software env, running codes (serial, shared- vs. distributed-memory, GPU, and their various permutations), efficient utilization, best practices
- CHAPEL
  - fast, modern programming language for both shared- and distributed-memory systems
  - high-level, easy-to-use abstractions for data and task parallelism
  - one syntax across 3 levels of hardware parallelism: multi-core CPUs, multi-node clusters, and GPUs
  - hides the distinction between multi-threading and multi-processing from the programmer
  - PGAS (Partitioned Global Address space) programming model + computation follows data
  - focus on data parallelism through two numerical examples: one embarrassingly parallel (2D Julia set fractals) and one tightly coupled (heat diffusion) ⇒ work on bottlenecks, load balancing, scaling
  - low-level interconnect communication hidden from the programmer (unlike in MPI)

Parallel coding (continued)





- julia
  - fast, high-level programming language for scientific computing and data science
  - just-in-time compilation + interactive shell
  - we take you on a journey to parallelize two problems: slow series and 2D Julia set
  - multi-threading with Base.Threads and ThreadsX.jl
  - multi-processing with Distributed.jl + arrays with DistributedArrays.jl and SharedArrays.jl
  - work on thread safety, race conditions, benchmarking, load balancing
- python™
  - aim: make it as fast + scalable as Julia or Chapel for numerical problems
  - explore ∼dozen different approaches: most fail for the slow series, and we analyze why
  - eventually settle on a combination of Numba for compilation and Ray for distributed computing
  - cover parallelizing with both Ray core and Ray data
  - our latest GPU modules: Arrays with CuPy, Numba JIT compilation to CUDA kernels, DataFrames with Polars and RAPIDS

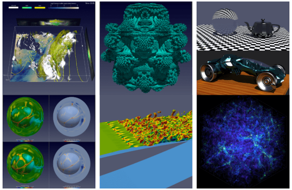
70 – 90 events each year across all formats

Current events at https://training.westdri.ca | Events | select the season

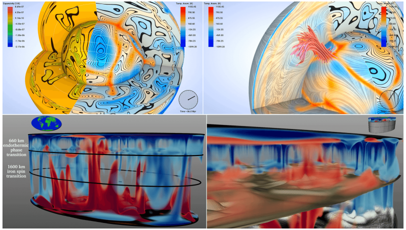
1. Biweekly webinars since March 2016
   - alternating between advanced and introductory topics; only topics not already covered in our courses
   - focus on new and/or impactful open-source tools and techniques that will benefit researchers
   - most webinars delivered by the two of us; last year partnered with UBC
   - ∼ 150 past webinars archived at https://training.westdri.ca (chronologically and by topic)
   - this term, webinars resumed in April to accommodate the Winter Visualization Series https://folio.vastcloud.org/winterseries
   - winter/spring 2026 training schedule https://training.westdri.ca/events/upcoming-training-winter-spring-2026
2. Weekly online courses since February 2023
3. Annual week-long summer schools since 2017
   - typically every June @SFU: 5 full days, in-person at the Big Data Hub, ∼ 70 registrants
   - ideally, we would run a couple per year and distribute them geographically; previously organized and taught these at UBC, UVic, U of Manitoba, U of Saskatchewan, U of Calgary, UNBC
   - this year June 1-5 @SFU (registration will open in April) ✆ tell us what topics you'd like us to cover
4. National and international training events: HSS/Visualization Winter Series, national courses, DHSI (since 2016), multi-year PIMS/M2PI collaboration, IHPCSS
5. Local workshops and bootcamps available upon request for departments and research groups

Researcher support and outreach

- Official support support@tech.alliancecan.ca from the Alliance, mostly for specific problems on our clusters $\Rightarrow$ tickets answered by HPC / ARC analysts across the country



- Followup from our training, support emails and tickets
- Visualization support
  - I lead the National Visualization Team `https://ccvis.netlify.app`
  - answer most visualization tickets in the national queue
  - HPC background $\Rightarrow$ can visualize very large (GBs to TBs and larger) datasets
  - recent sci-vis projects with researchers at UBC (in-situ visualization), UQAM (climate model visualization), UVic (climate CFD, optimization), UofA (engineering CFD)
  - happy to help with any visualization projects



- Near-annual *Visualize This!* contests
  - nationally since 2016 `https://ccvis.netlify.app/contests`
  - in 2021 hosted the international IEEE Vis Contest `https://scivis2021.netlify.app`
  - currently on pause: we have data and technical expertise; seeking collaborative outreach partners
- Subscribe to our weekly mailing list `https://training.westdri.ca/contact`

Opportunities for you

- We offer training on any research computing topic in any format mentioned earlier
  - we constantly develop new materials, can look at new topics and specific computational problems
  - couple of restrictions
    1. in general, we do not cover commercial (not open-source) tools
    2. we probably do not have specialized knowledge in your field $\Rightarrow$ with few exceptions, cannot teach domain-specific topics & tools, but we could develop a joint course with you (domain knowledge from you and computational expertise from us)
- We can teach guest lectures or collaborate on graduate-level university courses
- Email training@westdri.ca to request a workshop or suggest a topic
- Free to Canadian academic researchers

Numerical methods

Special interest in numerical methods
$\Rightarrow$ looking for collaboration on courses and materials

- computational fluid dynamics, conservation methods, Riemann solvers
- numerical radiative transfer (all regimes from free streaming to diffusion, 6D)
- serial / parallel ODE and PDE solvers, stiff systems and implicit methods
- adaptive techniques

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